



Assignment 3 - Mathematics-II
Departments: CSE,ECE,ENE
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📅 Apr 21, 2026 lecture
📍 M05, RSMS Building

1. Find the inverse Laplace transform of

(a) (i) $\frac{4p+15}{16p^2-25}$ (ii) $\frac{p-3}{p^2+2p+7}$

(b) (i) $\frac{p^2+1}{p^3+3p^2+2p}$ (ii) $\frac{p+1}{p^2+6p+25}$

(c) (i) $\frac{2p^2-4}{(p+1)(p-2)(p-3)}$ (ii) $\frac{p}{p^4+p^2+1}$

(d) (i) $\frac{2}{p} \left(1 + \frac{e^{-p}}{p}\right)$ (ii) $\frac{e^{-cp}}{p^2(p+a)}$, $c > 0$ (Use Second Shifting Theorem)

2. Find the inverse Laplace transform of (i) $\ln \frac{p(p+1)}{p^2+4}$ (ii) $\ln \left(1 + \frac{\alpha^2}{p^2}\right)$

3. Apply convolution theorem to evaluate: (i) $L^{-1} \left\{ \frac{p^2}{p^4-a^4} \right\}$ (ii) $L^{-1} \left\{ \frac{p}{(p^2+a^2)^3} \right\}$

4. Solve the following equations by Laplace transform

(a) $(D^3 - D^2 + 4D - 4)x = 68e^t \sin 2t; x = 1, Dx = -19, D^2x = -37$ at $t = 0$.

(b) $(D^3 - D^2 - D + 1)y = 8te^t; y(0) = 0, y'(0) = 1, y''(0) = 0$.

5. Prove that in the range $-\pi < x < \pi$, $\cosh ax = \frac{2a}{\pi} \sinh a\pi \left[\frac{1}{2a^2} + \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2+a^2} \cos nx \right]$

6. Find the Fourier series expansion for the function $x - x^3$ in the interval $-1 < x < 1$

7. Using the Fourier series expansion of $|x|$ in $(-\pi, \pi)$ show that

(i) $\sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} = \frac{\pi^4}{96}$ (ii) $\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$.

8. Expand $f(x) = x - \frac{x^2}{2}$ in $(0, 2)$ as Fourier sine series and hence evaluate $\sum_{n=1}^{\infty} \frac{1}{(2n-1)^6}$